GTIIT Spring 2025 Calculus 1M
Workshop 4

April 8, 2025

Try to solve the following exercises. Feel free to chat about them with your classmates, but make sure to write your own solutions. If you get stuck, just move on to the next one and come back later.

If you need any help at all —whether it's a tiny hint or a big clue— don't hesitate to ask me!

Important note: If you find any errors or typos, please let me know so I can correct them.

1. Calculate y':
   (a) $y = \tan(\frac{t}{t^2+1})$
   (b) $xe^y = y\sin(x)$
   (c) $y = e^{\cos x} + \cos(e^x)$
   (d) $y = \ln\sin(x) - \frac{1}{2}\sin(x^2)$
2. Use mathematical inducton (p. 72 Steart's book) to show that if $f(x) = xe^x$, then $f^{(n)}(x) = (x+n)e^x$.
3. Suppose that:

$$f(1) = 2, \quad f'(1) = 3, \quad f(2) = 3, \quad f'(2) = 2 \quad g(1) = 3, \quad g'(1) = 1, \quad g(2) = 1, \quad g'(2) = 4,$$

   (a) If $S(x) = f(x) + g(x)$, find $S'(1)$.
   (b) If $P(x) = f(x)g(x)$, find $P'(2)$.
   (c) If $Q(x) = \frac{f(x)}{g(x)}$, find $Q'(1)$.
   (d) If $C(x) = fg(x)$, find $C'(2)$.
4. (a) Find and equation of the tangent line to the curve $y = e^x$ that is parallel to the line $1 = x - 4y$.
   (b) Find and equation of the tangent line to the curve $y = e^x$ that passes through the origin.
   (c) Find f' in terms of g' where $f(x) = \ln|g(x)|$.
   (d) Find h' in terms of f' and g' where $h(x) = \frac{f(x)g(x)}{f(x)+g(x)}$.